Science Team Weekly

A coherent science-literacy cycle: analyze sources, make thinking visible, respond to evidence, and revise.

THIS WEEK'S DEPARTMENT LEARNING ARC

From CUBS Analysis to Feedback, Revision, and the Next Learning Experience

This week's PLCs and Wednesday PD moved through one shared lesson-internalization cycle. Teachers experienced a science task as learners, used evidence to plan rigorous instruction, practiced student-facing literacy routines, and then read completed student writing to choose a precise next step.

MONDAY · EXPERIENCE

Teachers completed a shared writing benchmark as students, used **CUBS** to analyze the prompt and visuals, and color-coded Claim, Evidence, and Reasoning.

TUESDAY · PLAN

Teams used anticipated responses to map a lesson arc, protect grade-level thinking, and plan a feedback prompt plus a student revision action.

FRIDAY · ADJUST

Teachers read 3–5 writing samples against a rubric, named one priority need, wrote actionable feedback, and selected one evidence check.

WEDNESDAY PD · STRENGTHENING SCIENCE LITERACY

CUBS + Color-Coded CER: A Student-Facing Routine

During the October 28 PD, teachers adapted CUBS and color-coded CER to authentic course sources, visuals, data displays, models, investigations, and writing tasks. The goal was not an additional grading system. It was a shared way for students to slow down, identify usable information, explain their thinking, receive feedback, and revise.

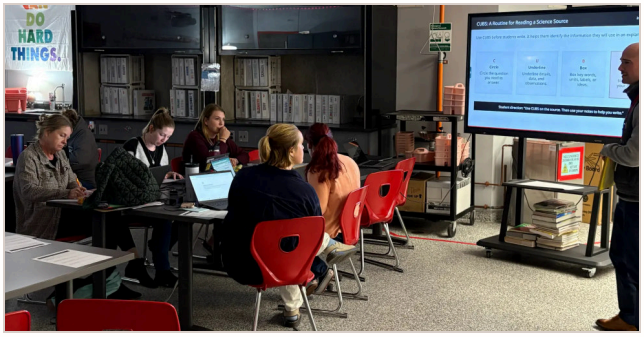
STUDENT ROUTINE

CUBS: Circle the central question, Underline relevant details or data, Box key vocabulary, and Summarize the science idea.

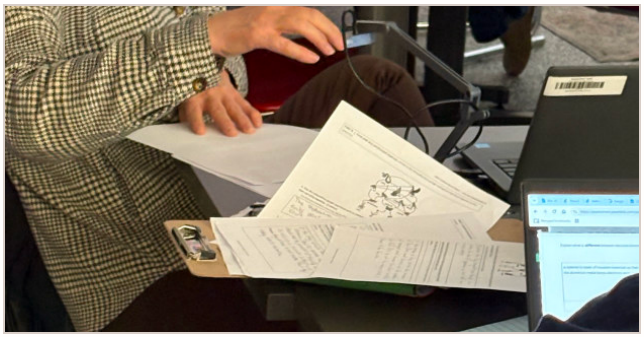
CER: Make the Claim, Evidence, and Reasoning visible so students and teachers can identify the next revision move.

THANK YOU

Thank you to **Mr. Tyler Gilbert** and **Ms. Seannine Tveit** for leading our Science Literacy PD this Wednesday and helping colleagues rehearse a practical routine for source analysis, evidence, feedback, and revision.



Science Literacy PD: *teachers engage with a student-facing CUBS routine before adapting it to their own courses.*



Planning from evidence: *the work connects task analysis to feedback and the next student action.*

Thank you, Science Team: *We appreciate everyone's strong attendance and attention in PLC and PD, along with the willingness to take risks, try new strategies, and collaborate with colleagues to strengthen classroom instruction.*

WHY THIS CYCLE MATTERS

When students read a source, identify evidence, construct a response, revise from feedback, and show new evidence of learning, science literacy becomes part of the learning experience—not an add-on.

WHAT TEAMS CARRY FORWARD

STUDENT DIRECTION

One clear CUBS or CER direction students can use independently.

EVIDENCE LOOK-FOR

One student-work pattern that will show whether the strategy is working.

REVISION MOVE

One feedback prompt and one specific action students will take next.

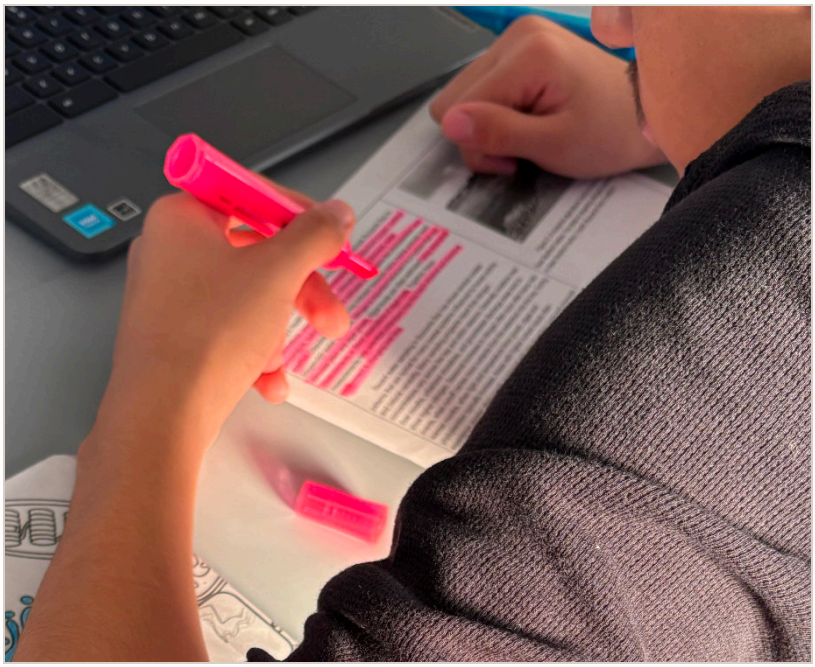
Students Make Their Thinking Visible

Color-coded CER moves from professional learning into classroom practice.

FROM PD STRATEGY TO STUDENT ACTION

Reading Closely, Selecting Evidence, and Explaining Why

Students are beginning to apply the color-coded CER routine to science articles and written responses. The strategy helps students distinguish a conclusion from the evidence that supports it—and identify where reasoning is needed to connect the two.



Science literacy in action: *a student uses color coding to locate information and prepare an evidence-based explanation.*

THANK YOU, SCIENCE TEACHERS

We appreciate teachers who are incorporating the strategies from PLC and PD learning into daily instruction and giving students repeated opportunities to analyze, explain, and revise.

A CLEAR PATH FROM SOURCE TO RESPONSE

What Color Coding Can Reveal

CLAIM
Students identify the answer, prediction, or conclusion that responds to the scientific question.

EVIDENCE
Students mark relevant details from the article, data display, model, or investigation—not every fact on the page.

REASONING
Students explain why the chosen evidence supports the claim using a scientific idea, pattern, or mechanism.

REVISION
Feedback asks students to revisit the source, select or clarify evidence, and improve one specific part of the response.

STUDENT SELF-CHECK BEFORE SUBMITTING

Can I point to my claim? Does it answer the question?

Can I show my evidence? Is it relevant and accurate?

Can I explain my reasoning? Did I say why the evidence supports the claim?

SUPPORT THAT PRESERVES RIGOR

CUBS and color coding are not shortcuts to the answer. They help students access the source and make the science thinking they must construct visible to themselves, peers, and teachers.

Instructional focus: *Students do not just highlight. They use annotations to select evidence, talk through an idea, and strengthen a scientific explanation.*

THE LEARNING TRANSFER

The strongest sign of professional learning is student learning. This week, teachers' shared work on CUBS and CER is already helping students read and respond to science sources with more purpose.

FEEDBACK THAT LEADS TO REVISION

1. NAME THE CRITERION
"Your response has a claim, but the reasoning needs to explain why the evidence supports it."

2. SEND STUDENTS BACK
"Look again at the graph, article, or model. Which detail helps explain the pattern?"

3. ASK FOR ONE ACTION
"Add or revise one sentence that links your evidence to the scientific idea."

TEACHER FEEDBACK FRAME

"You need to ___. Look at ___. Add or revise ___." This keeps feedback specific, usable, and connected to student action.

Field Science, Habitat Stewardship, and Innovation

AP Environmental Science students investigate invasive phragmites at Cuttyhunk Island.

CUTTYHUNK ISLAND FIELD STUDY

From Removal Challenge to Sustainable Resource Question

Led by **Ms. Michelle Carvalho**, AP Environmental Science students traveled to Cuttyhunk Island for hands-on field work focused on phragmites. The experience connected habitat stewardship, ecological data collection, problem solving, and career exploration.



Boots-on-the-ground science: *students work in a coastal habitat while examining the scale and persistence of invasive phragmites.*



From field collection to analysis: *students examine the material and consider environmental questions it raises.*

A REAL ENVIRONMENTAL CHALLENGE

Why Phragmites Calls for Persistent Problem Solving

Removing invasive phragmites requires significant time, physical effort, coordination, and repeated work. The challenge gives students an honest view of environmental restoration: ecological problems are often complex, labor intensive, and not solved by a single intervention.

Students considered a second question alongside removal: **Can phragmites become a resource material?** Reframing an invasive plant as a possible material for innovation invites students to examine sustainable solutions, not simply identify a problem.

FIELD-SCIENCE THINKING

Observe: What is the plant doing to the habitat?

Collect: What data or material will help answer the question?

Innovate: How could reuse, design, or art extend the impact of removal work?

CAREER CONNECTION

For students considering environmental studies, the field study offered a look at the work that can happen beyond the classroom: habitat assessment, data collection, restoration, communication, and collaborative problem solving.

LOOKING AHEAD

A follow-up learning experience can revisit habitat impacts, analyze the data students collected and why it matters, and extend the art-and-innovation component of the project.

THANK YOU

We are grateful to **Ms. Carvalho** for creating a place-based field experience that brings environmental science, sustainability, and student curiosity together.

Big question: *How can environmental science move from identifying an invasive-species problem to designing a long-term, sustainable response?*

SCIENCE BEYOND THE CLASSROOM

Field studies help students see that evidence, persistence, and creative thinking all matter when a community faces an environmental challenge.

FROM FIELDWORK TO FUTURE LEARNING

HABITAT IMPACT

Use observations to ask how phragmites can shape light, space, and habitat conditions in a coastal environment.

DATA STORY

Share what students collected, explain why each data point matters, and identify the questions the evidence cannot answer yet.

INNOVATION CHALLENGE

Evaluate how a resource or art-design idea could complement removal efforts and support a sustainable response.

Modeling Osmosis: From Solute Patterns to Water Movement

Students use visual models, academic talk, and evidence to explain tonicity and cell transport.

PRE-AP BIOLOGY · MS. GRACE STEPHENSON

Card Sorts Turn a Cell Model into a Scientific Explanation

Students used cell-model cards to examine relative solute concentrations and sort scenarios into **hypotonic, isotonic,** and **hypertonic** environments. The task asked students to use a model as evidence—not simply memorize vocabulary.



Model-based reasoning: *students compare solute concentrations inside and outside a cell before classifying the environment.*

VOCABULARY WITH PURPOSE

Hypotonic, isotonic, hypertonic, solute, solvent, selectively permeable membrane, and **osmosis** became tools for explaining what the model showed.

WHAT STUDENTS WERE ASKED TO DO

Analyze, Discuss, Justify, Revise

ANALYZE THE MODEL
Compare the number of solute particles inside and outside the cell and notice the concentration difference.

CLASSIFY THE ENVIRONMENT
Use the evidence in the model to decide whether the surrounding solution is hypotonic, isotonic, or hypertonic.

PREDICT WATER MOVEMENT
Explain the direction water will move across a selectively permeable membrane and describe the likely effect on the cell.

REVISE THINKING
Use questions and feedback to revisit a classification or prediction and strengthen the explanation.

TALK MOVE FOR SCIENTIFIC REASONING

"I classified this model as ___ because the solute concentration is higher/lower ___. Water will move ___, which may cause the cell to ___."

Why a card sort matters: *students must compare representations, defend a decision, listen to peers, and connect visual evidence to a biological mechanism.*

EVIDENCE
Use the model's solute pattern as the basis for the explanation.

LANGUAGE
Use precise vocabulary to describe tonicity and water movement.

REVISION
Change a placement or explanation when new reasoning makes the pattern clearer.

STUDENT LEARNING SPOTLIGHT

Thank you to **Ms. Grace Stephenson** for creating a learning experience where students use visual models, discussion, and evidence to make sense of osmosis and cell membrane transport.

A STUDENT REASONING CHECK

HYPOTONIC
Compare the outside and inside solute levels. Predict water movement and describe what may happen to the cell.

ISOTONIC
Look for balance. Explain why water can move in both directions without a net change in cell volume.

HYPERTONIC
Use the concentration pattern to explain why water may leave a cell and how that affects its condition.

EVIDENCE QUESTION

"What feature in the model helped you decide?" Asking for the visual evidence moves students beyond a vocabulary label to a scientific explanation.

Tiny RNA, Big Control

How small RNAs help tune protein production—and why "junk DNA" is an outdated shortcut.

SCIENCE CORNER · MOLECULAR BIOLOGY

Small RNAs Do Not Make Proteins. They Help Decide How Many Get Made.

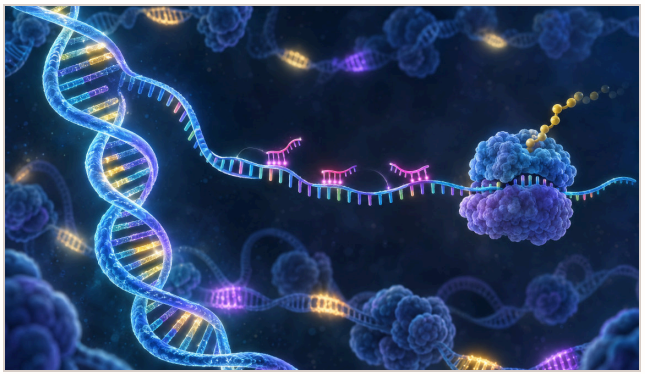
A cell's DNA contains instructions, but instructions are not automatically turned into proteins. Many layers of regulation determine which messenger RNAs are used and how much protein is produced. MicroRNAs are one important layer.

MicroRNAs are tiny, non-protein-coding RNA molecules. A mature microRNA can pair with complementary sequences on a messenger RNA (mRNA), often as part of a protein-containing silencing complex. That pairing can reduce translation—the process by which a ribosome reads mRNA to build a protein—or can help trigger mRNA decay. In other words, microRNAs regulate **protein output** by acting on the mRNA message, rather than by directly changing a finished protein [1–3].

The importance of this mechanism became widely recognized after the discovery of the *lin-4* small RNA in a tiny roundworm. In 1993, researchers showed that this RNA did not code for a protein; instead, its sequence was complementary to part of an mRNA and was linked to reduced production of the LIN-14 protein [2]. In 2024, Victor Ambros and Gary Ruvkun received the Nobel Prize in Physiology or Medicine for the discovery of microRNA and its role in post-transcriptional gene regulation [1].

GENE REGULATION IN ONE PATHWAY

DNA → mRNA → protein. MicroRNAs add a control point between mRNA and protein, helping cells fine-tune which instructions are translated and how strongly.



Conceptual view: *small RNAs pair with mRNA targets and can reduce translation or promote mRNA decay.*

NETWORK EFFECT

A single microRNA can influence multiple mRNAs, and an individual mRNA can be regulated by more than one microRNA. This makes gene regulation a network—not a single switch [1].

WHAT ABOUT "JUNK DNA"?

Noncoding Does Not Mean Useless

Only about 1% of human DNA directly codes for proteins. Earlier discussions sometimes called the rest "junk DNA," but that phrase is too simple. Many noncoding regions contain regulatory sequences—such as promoters, enhancers, and silencers—that help determine when and where genes are active. Other noncoding regions provide instructions for RNAs, including microRNAs [4].

Scientists should still be precise: not every noncoding sequence has a known function, and researchers continue to test what different regions do. Large genome projects have mapped many regulatory and transcribed elements outside protein-coding genes, showing that the noncoding genome is an active area of discovery [5].

WHY THIS MATTERS FOR HEALTH

Gene regulation must be coordinated during development and in adult tissues. When regulatory systems are disrupted, including microRNA pathways, cells may produce proteins at the wrong time or in the wrong amount. Such changes can contribute to disease, including cancer, but they are part of a complex, multifactorial process—not a single-cause explanation [1, 3].

SCIENCE-LITERACY CONNECTION

Try tracing the evidence chain: **noncoding DNA → microRNA → mRNA target → protein output → cell behavior.** This is an example of how a small molecular change can affect a larger biological system.

TAKEAWAY

The phrase "small RNA" can be misleading. MicroRNAs are tiny, but their ability to coordinate gene-expression networks gives them a powerful role in how cells develop, respond, and maintain balance.

SCIENCE CORNER VOCABULARY + THINKING TOOL

FOUR TERMS TO KNOW

MicroRNA: a short noncoding RNA that can regulate a target mRNA.

mRNA: a molecular message that can be translated into a protein.

Translation: the ribosome-guided process of building a protein from an mRNA sequence.

Noncoding DNA: DNA that does not directly encode proteins; some regions regulate gene activity or produce functional RNAs.

TRACE THE SYSTEM

1. A noncoding DNA region is transcribed into a microRNA.

2. The microRNA pairs with a matching region on an mRNA.

3. Translation is reduced or the mRNA is more likely to be degraded.

4. The cell's protein output changes—one reason small RNAs can influence whole gene networks.

SELECTED REFERENCES

[1] Nobel Assembly at Karolinska Institutet. *The Nobel Prize in Physiology or Medicine 2024: Press Release.*

[2] Lee, R. C., Feinbaum, R. L., & Ambros, V. (1993). *Cell*, 75, 843–854. The *lin-4* gene encodes small RNAs complementary to *lin-14*.

[3] Valencia-Sanchez, M. A., Liu, J., Hannon, G. J., & Parker, R. (2006). *Genes & Development*, 20, 515–524. Control of translation and mRNA degradation by miRNAs and siRNAs.

[4] MedlinePlus Genetics, NIH. *What is noncoding DNA?* [5] ENCODE Project Consortium (2012). *Nature*, 489, 57–74. An integrated encyclopedia of DNA elements in the

Language Supports That Keep Science Thinking Rigorous

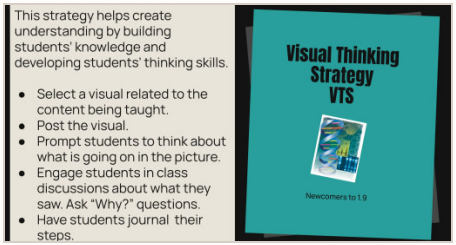
Practical strategies that help multilingual learners access, discuss, and explain grade-level science.

SHARED DEPARTMENT RESOURCE

Thank You, Ms. Hope Bynoe and Ms. Brittany Jenney

We appreciate **Ms. Hope Bynoe** and **Ms. Brittany Jenney** for sharing instructional strategies that support English learners while strengthening learning for all science students. The common purpose is clear: make the language of science visible without reducing the scientific thinking students are asked to do.

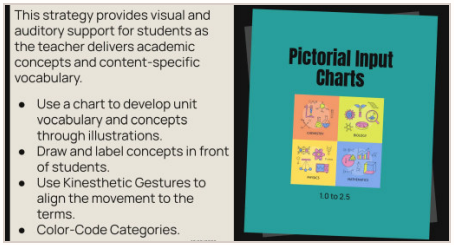
Department appreciation: Special thanks to our custodial team for distributing instructional materials across classrooms, and to **Ms. Andrea E. Garcelon-Ramos** for ordering the instructional materials and equipment that make our science learning possible.



Visual Thinking Strategy: *use an image to prompt observation and discussion.*

VISUAL THINKING

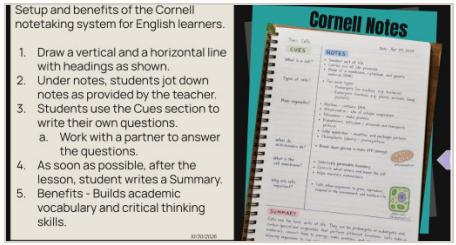
Choose a content-connected visual, ask students what is happening and why, then invite students to journal their thinking. The routine builds knowledge, talk, and evidence-based noticing.



Pictorial Input Charts: *concepts and vocabulary through drawings, labels, and gestures.*

PICTORIAL INPUT

Develop unit vocabulary with illustrations, labels, color-coded categories, and gestures. Drawing key ideas in front of students helps connect a word to a scientific meaning.



Cornell Notes: *organize information, questions, and a summary.*

CORNELL NOTES

Students record notes, write their own cues or questions, discuss with a partner, and summarize. This structure builds academic vocabulary and helps students revisit a concept.

FIVE PRACTICAL SUPPORTS FOR MULTILINGUAL LEARNERS

- Use proficiency levels to plan the right level of language support.
- Integrate listening, speaking, reading, and writing with the science content.
- Slow the delivery of key ideas and limit idioms that can hide meaning.
- Teach the essential concepts with clear, manageable language.
- Use visuals consistently to connect academic vocabulary to an idea or phenomenon.

CONNECTION TO THIS WEEK'S LITERACY CYCLE

SUPPORT THE SAME RIGOROUS TASK

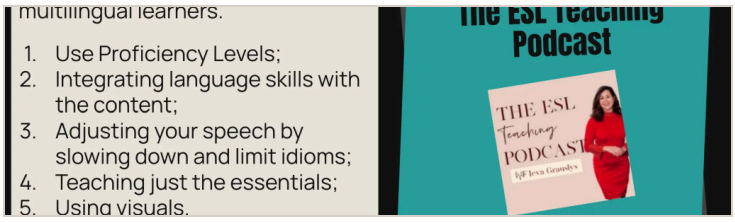
CUBS, color-coded CER, pictorial input, visual thinking, and Cornell Notes can all give students a path into a science source, model, graph, or writing task. The supports clarify the language and process while students remain responsible for analyzing evidence and constructing an explanation.

PLANNING QUESTION

Before the lesson, ask: "What will students see, say, read, write, and revise—and what support will help them do each part independently?"

DEPARTMENT COMMITMENT

Across our Science Department, we will continue to build learning experiences where every student can read science, discuss science, use evidence, and communicate an explanation with confidence.



Shared ELD resource: *language supports can be planned alongside the science task.*

TURN ONE STRATEGY INTO A DAILY ROUTINE

Before: preview a visual and 2–3 key words. **During:** give students a structured way to speak, annotate, or record notes. **After:** ask for a brief spoken or written explanation that returns to the science evidence.

The goal is not to simplify the science. The goal is to make the path into the science more visible.

READY-TO-USE SCIENCE TALK FRAMES

NOTICE

"Based on the model, graph, or text, I notice ___."

EXPLAIN

"The evidence suggests ___ because ___."

BUILD ON

"I agree/disagree with ___ because I can add ___."

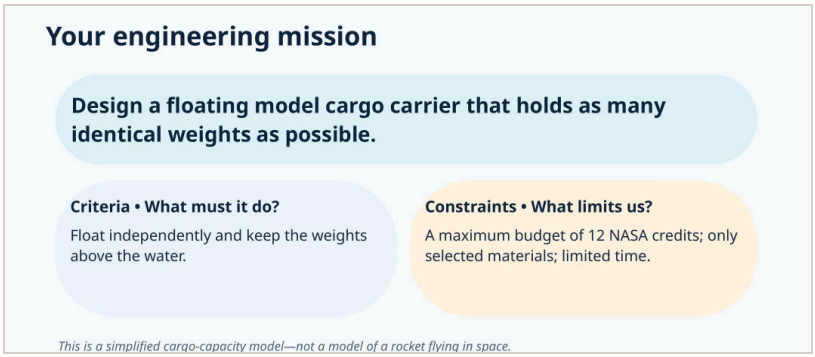
Engineering Under Constraints: The NASA Payload Challenge

Ms. Cassidy Barrett brings the Artemis II context to life through design, data, evidence, and revision.

ARTEMIS II · ENGINEERING DESIGN IN ACTION

From Mission Constraints to Student-Made Solutions

Ms. Cassidy Barrett is implementing the NASA Payload Engineering Challenge, a hands-on learning experience in which students design a floating cargo carrier, make careful material decisions, test a prototype, and use evidence to explain and improve their solution.



Mission context: *students connect a classroom engineering model to the resource limits engineers must consider on long-duration space missions.*

THE DESIGN BRIEF

Mission: Design a floating model cargo carrier that holds the greatest number of identical weights while staying within a materials budget.

Driving question: How can we find, make, and recycle the substances we need to live on and beyond Earth?

A FULL ENGINEERING CYCLE

Plan, Build, Test, Explain, Improve

PLAN WITH CRITERIA AND CONSTRAINTS

Students select from approved materials, stay within a maximum budget of **12 NASA credits**, sketch a design, and predict performance.

BUILD A PROTOTYPE

Teams make a carrier that can float independently, keep the payload above water, and remain stable as identical weights are added.

TEST FAIRLY AND RECORD DATA

Students add weights one at a time, record credits spent and maximum payload, and document how the model succeeds or fails.

USE EVIDENCE TO REDESIGN

Students evaluate trade-offs, explain a design choice with evidence, and propose a specific improvement for a new test.

ENGINEERING TALK FRAME

"Our design was successful/partly successful because ____. We know this because it held ____ weights and cost ____ credits. Next, we would change ____ because ____."

Why the challenge matters: *Students experience that strong engineering solutions are not simply the ones that hold the most weight—they are solutions evaluated with evidence, criteria, constraints, and trade-offs.*

WHAT STUDENTS MAKE VISIBLE

CRITERIA

Does the model float, keep the payload above water, and carry weights stably?

CONSTRAINTS

Which limited materials and budget choices shaped the design?

EVIDENCE

What do the test results show about payload capacity, stability, efficiency, and needed revisions?

- Which design held the most weights? What is your evidence?
- Which design used its credits most efficiently?
- What trade-offs did you notice between cost and performance?
- How would you improve your design if you tested again?

From data to explanation: *students compare results, defend a conclusion, and name a trade-off between cost and performance.*

LANGUAGE SUPPORTS FOR ENGINEERING

Students use the academic vocabulary **payload, criteria, constraints, resources,** and **trade-offs** as tools for communicating a design decision—not as isolated words to memorize.

The challenge aligns with Massachusetts HS-ETS1-3: evaluate solutions using criteria, constraints, and trade-offs, including cost, safety, and reliability.

CLASSROOM SPOTLIGHT

Thank you to **Ms. Cassidy Barrett** for implementing a high-quality engineering experience that invites students to problem-solve, analyze data, collaborate, and use evidence to improve a design.

FROM STUDENT WORK TO THE NEXT DESIGN MOVE

EVIDENCE REVIEW: WHAT TO LOOK FOR

- Does the student name a criterion and a constraint?
- Does the explanation cite a measurement, observation, or comparison?
- Does the student identify a meaningful trade-off?
- Does the redesign respond to evidence rather than a guess?

FEEDBACK THAT PROMPTS ACTION

"Your data show ____. Explain how that evidence supports your claim about the design."

"You named a trade-off. What specific change would you test next, and why?"

This feedback keeps students focused on the evidence-to-revision cycle that engineers use to improve solutions.